KL Divergence Between Gaussians

the general form

Introduction to Generative AI in Computer Vision

Supplementary note — optional further reading

Why this note. The companion note ("the case we use in the lectures") covers everything the course needs, because every Gaussian appearing in the lectures has a diagonal covariance matrix. This note removes that restriction and allows arbitrary covariances.

Two reasons to read it. First, the general formula is what you will meet in papers, written with a trace and a log-determinant, and it is worth being able to recognise it. Second, the proof introduces one small trick — turning the expectation of a quadratic form into a trace — that reappears throughout machine learning whenever second moments are involved.

Nothing in the lectures depends on this note. Its results specialise back to the companion note's, which we verify in Section 4.

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1 Statement

Proposition 1. *Let $q = \mathcal{N}(\boldsymbol{\mu}_q, \boldsymbol{\Sigma}_q)$ and $p = \mathcal{N}(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$ be Gaussian distributions on $\mathbb{R}^d$ with positive definite covariances. Then*

$$\mathcal{D}_{\mathrm{KL}}(q \,\|\, p) = \frac{1}{2}\left[\operatorname{tr}\!\left(\boldsymbol{\Sigma}_p^{-1}\boldsymbol{\Sigma}_q\right) + (\boldsymbol{\mu}_p - \boldsymbol{\mu}_q)^\top \boldsymbol{\Sigma}_p^{-1}(\boldsymbol{\mu}_p - \boldsymbol{\mu}_q) - d + \log\frac{\det \boldsymbol{\Sigma}_p}{\det \boldsymbol{\Sigma}_q}\right]. \tag{1}$$

As always we are computing

$$\mathcal{D}_{\mathrm{KL}}(q \,\|\, p) = \int q(\mathbf{x}) \log\frac{q(\mathbf{x})}{p(\mathbf{x})}\,\mathrm{d}\mathbf{x} = \mathbb{E}_{\mathbf{x}\sim q}\left[\log\frac{q(\mathbf{x})}{p(\mathbf{x})}\right], \tag{2}$$

the average *under q* of the log-ratio of the two densities.

2 The one tool we need

In one dimension the only facts required were $\mathbb{E}[(x-\mu_q)^2] = \sigma_q^2$ and $\mathbb{E}[x-\mu_q] = 0$. In d dimensions the squares become quadratic forms $\mathbf{a}^\top\mathbf{M}\mathbf{a}$, and the following identity is what lets us average them.

Lemma 1. *Let $\mathbf{a}$ be a random vector in $\mathbb{R}^d$ and let $\mathbf{M} \in \mathbb{R}^{d\times d}$ be a fixed matrix. Then*

$$\mathbb{E}\Big[\mathbf{a}^\top\mathbf{M}\mathbf{a}\Big] = \operatorname{tr}\Big(\mathbf{M}\,\mathbb{E}\Big[\mathbf{a}\mathbf{a}^\top\Big]\Big). \tag{3}$$

Proof. The quantity $\mathbf{a}^\top\mathbf{M}\mathbf{a}$ is a scalar, and a scalar equals the trace of itself, so $\mathbf{a}^\top\mathbf{M}\mathbf{a} = \operatorname{tr}(\mathbf{a}^\top\mathbf{M}\mathbf{a})$. The trace is unchanged by cyclic permutation of a product, which moves $\mathbf{a}^\top$ to the back: $\operatorname{tr}(\mathbf{a}^\top\mathbf{M}\mathbf{a}) = \operatorname{tr}(\mathbf{M}\mathbf{a}\mathbf{a}^\top)$. Finally, a trace is a finite sum of matrix entries, so it commutes with the expectation. □

The value of (3) is that the randomness is now confined to the single matrix $\mathbb{E}[\mathbf{a}\mathbf{a}^\top]$ — and for a Gaussian centred at $\boldsymbol{\mu}_q$, with $\mathbf{a} = \mathbf{x} - \boldsymbol{\mu}_q$, that matrix is exactly the covariance $\boldsymbol{\Sigma}_q$.

3 Proof of Proposition 1

Step 1: the log-density. For any Gaussian,

$$\log \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = -\frac{1}{2}\Big[d \log 2\pi + \log\det \boldsymbol{\Sigma} + (\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big]. \tag{4}$$

This is the d-dimensional counterpart of $-\frac{1}{2}\log(2\pi\sigma^2) - (x-\mu)^2/(2\sigma^2)$: the variance is replaced by a determinant, and the square by a quadratic form.

Step 2: subtract. Apply (4) to q and to p and subtract. The two $d\log 2\pi$ terms are identical and cancel:

$$\log \frac{q(\mathbf{x})}{p(\mathbf{x})} = \frac{1}{2}\log\frac{\det \boldsymbol{\Sigma}_p}{\det \boldsymbol{\Sigma}_q} - \frac{1}{2}T_q(\mathbf{x}) + \frac{1}{2}T_p(\mathbf{x}), \tag{5}$$

where

$$T_q(\mathbf{x}) := (\mathbf{x}-\boldsymbol{\mu}_q)^\top \boldsymbol{\Sigma}_q^{-1}(\mathbf{x}-\boldsymbol{\mu}_q), \qquad T_p(\mathbf{x}) := (\mathbf{x}-\boldsymbol{\mu}_p)^\top \boldsymbol{\Sigma}_p^{-1}(\mathbf{x}-\boldsymbol{\mu}_p).$$

By (2) only $\mathbb{E}_q[T_q]$ and $\mathbb{E}_q[T_p]$ remain to be computed; the log-determinant term does not involve $\mathbf{x}$ and passes straight through.

Step 3: the form matched to its own Gaussian. Put $\mathbf{a} = \mathbf{x} - \boldsymbol{\mu}_q$ in Lemma 1. Since $\mathbf{x} \sim q$ has mean $\boldsymbol{\mu}_q$, we have $\mathbb{E}_q[\mathbf{a}\mathbf{a}^\top] = \boldsymbol{\Sigma}_q$, so

$$\mathbb{E}_q[T_q] = \operatorname{tr}\big(\boldsymbol{\Sigma}_q^{-1}\boldsymbol{\Sigma}_q\big) = \operatorname{tr}(\mathbf{I}_d) = d. \tag{6}$$

This is the origin of the $-d$ in (1), and note that it depends on neither $\boldsymbol{\mu}_q$ nor $\boldsymbol{\Sigma}_q$. It is the multivariate version of the $-\frac{1}{2}$ in the scalar formula.

Step 4: the mismatched form. Now $\mathbf{x}$ is distributed around $\boldsymbol{\mu}_q$ while the quadratic form is centred at $\boldsymbol{\mu}_p$. Exactly as in the scalar proof, split the deviation so the random part is centred:

$$\mathbf{x} - \boldsymbol{\mu}_p = \underbrace{(\mathbf{x}-\boldsymbol{\mu}_q)}_{\text{random, mean } \mathbf{0}} + \underbrace{(\boldsymbol{\mu}_q - \boldsymbol{\mu}_p)}_{\text{deterministic}}.$$

Because $\boldsymbol{\Sigma}_p^{-1}$ is symmetric, expanding gives three pieces,

$$T_p = (\mathbf{x}-\boldsymbol{\mu}_q)^\top \boldsymbol{\Sigma}_p^{-1}(\mathbf{x}-\boldsymbol{\mu}_q) + 2(\boldsymbol{\mu}_q-\boldsymbol{\mu}_p)^\top \boldsymbol{\Sigma}_p^{-1}(\mathbf{x}-\boldsymbol{\mu}_q) + (\boldsymbol{\mu}_q-\boldsymbol{\mu}_p)^\top \boldsymbol{\Sigma}_p^{-1}(\boldsymbol{\mu}_q-\boldsymbol{\mu}_p),$$

which we average under q one at a time:

- the first is Lemma 1 again, now with $\mathbf{M} = \boldsymbol{\Sigma}_p^{-1}$, giving $\operatorname{tr}(\boldsymbol{\Sigma}_p^{-1}\boldsymbol{\Sigma}_q)$;
- the second vanishes: it is linear in $\mathbf{x} - \boldsymbol{\mu}_q$, and $\mathbb{E}_q[\mathbf{x} - \boldsymbol{\mu}_q] = \mathbf{0}$;
- the third is already deterministic.

Therefore

$$\mathbb{E}_q[T_p] = \operatorname{tr}\big(\boldsymbol{\Sigma}_p^{-1}\boldsymbol{\Sigma}_q\big) + (\boldsymbol{\mu}_q-\boldsymbol{\mu}_p)^\top \boldsymbol{\Sigma}_p^{-1}(\boldsymbol{\mu}_q-\boldsymbol{\mu}_p). \tag{7}$$

This is the matrix version of “variance plus squared bias”.

Step 5: collect. Insert (6) and (7) into the expectation of (5):

$$\mathcal{D}_{\mathrm{KL}}(q \,\|\, p) = \frac{1}{2} \log \frac{\det \boldsymbol{\Sigma}_p}{\det \boldsymbol{\Sigma}_q} - \frac{1}{2} d + \frac{1}{2} \operatorname{tr}\bigl(\boldsymbol{\Sigma}_p^{-1} \boldsymbol{\Sigma}_q\bigr) + \frac{1}{2} (\boldsymbol{\mu}_q - \boldsymbol{\mu}_p)^\top \boldsymbol{\Sigma}_p^{-1} (\boldsymbol{\mu}_q - \boldsymbol{\mu}_p),$$

which is (1). The quadratic form is unaffected by writing $\boldsymbol{\mu}_p - \boldsymbol{\mu}_q$ instead of $\boldsymbol{\mu}_q - \boldsymbol{\mu}_p$, since $(-\mathbf{v})^\top \mathbf{M}(-\mathbf{v}) = \mathbf{v}^\top \mathbf{M} \mathbf{v}$. □

4 Consistency with the companion note

As a check, (1) must reproduce the scalar formula. Take $d = 1$, so $\boldsymbol{\Sigma}_q = \sigma_q^2$ and $\boldsymbol{\Sigma}_p = \sigma_p^2$ are numbers: the trace is σ_q^2/σ_p^2, the quadratic form is $(\mu_q - \mu_p)^2/\sigma_p^2$, we subtract $d = 1$, and the log-determinant ratio is $\log(\sigma_p^2/\sigma_q^2) = 2 \log(\sigma_p/\sigma_q)$. Hence

$$\mathcal{D}_{\mathrm{KL}}(q \| p) = \frac{1}{2} \left[\frac{\sigma_q^2 + (\mu_q - \mu_p)^2}{\sigma_p^2} - 1 + 2 \log \frac{\sigma_p}{\sigma_q} \right] = \log \frac{\sigma_p}{\sigma_q} + \frac{\sigma_q^2 + (\mu_q - \mu_p)^2}{2\sigma_p^2} - \frac{1}{2},$$

which is exactly the scalar result of the companion note. Combined with the additivity lemma proved there, both corollaries used in the lectures follow again.

Remark. Nothing in this proof used the diagonality of either covariance, so it also covers correlated Gaussians. The price is the machinery: a determinant, a matrix inverse, and Lemma 1. That is why the companion note avoids all three — for diagonal covariances they are simply not needed.